Cross-Modal Fact-Checking: Detecting Out-of-Context Multimodal Misinformation using Contrastive Learning and Multi-Head Authenticity Detection

Wasikul Hasan Fahim
Dept. of Computer Science and Engineering
University of Asia Pacific
Dhaka, Bangladesh
Student ID: 22201037
22201037@uap-bd.edu

Md. Akif Hossain
Dept. of Computer Science and Engineering
University of Asia Pacific
Dhaka, Bangladesh
Student ID: 22201029
22201029@uap-bd.edu

Naima Tabassum Antora
Dept. of Computer Science and Engineering
University of Asia Pacific
Dhaka, Bangladesh
Student ID: 22201068
22201068@uap-bd.edu

***Abstract*—The rapid proliferation of digital media has significantly increased the spread of out-of-context misinformation, where authentic images are paired with deceptive textual claims. Traditional single-modality fact-checking systems often fail to identify the subtle semantic inconsistencies between visual and textual content. This project develops a deep learning based cross-modal fact-checking system, named the Cross-Modal Fact-Checker, designed to detect mismatches between an image and its associated claim. The proposed approach integrates a Contrastive Language-Image Pre-training (CLIP) visual encoder and a Bidirectional Encoder Representations from Transformers (BERT) text encoder. The extracted features are projected into a shared 512 dimensional embedding space, concatenated, and passed through a fact-checking head that produces a match probability. A semantic cosine similarity score between the two embeddings is also reported as an interpretable alignment signal. The model is trained on a balanced manifest derived from COCO, Flickr30k, NewsCLIPpings and CIFAKE. Training is carried out using the PyTorch framework with the AdamW optimizer, a batch size of 16 and a learning rate of 1e-4. The experimental evaluation on a 200 sample held-out test snapshot shows that the model attains a ROC-AUC of 0.98, with a fake-class recall of 0.99 and a real-class precision of 0.97 at the operating threshold of 0.77. These results demonstrate the effectiveness of jointly learned visual-semantic embedding spaces in reducing multimodal disinformation across digital communication platforms.**

***Index Terms*—Multimodal learning, misinformation detection, cross-modal fact-checking, CLIP-BERT fusion, contrastive learning, natural language processing, computer vision.**

I. Introduction

A. Background

Social media platforms have become a primary channel through which news, opinion and visual content circulate. Along with this convenience, however, comes a sharp rise in multimodal misinformation. The most common form of such misinformation pairs an authentic image with a misleading or out-of-context textual claim; for instance, an old disaster image may be re-shared with a caption claiming that the event just occurred at a different location. This phenomenon is referred to in the literature as out-of-context misinformation [1].

B. Motivation

Out-of-context misinformation is particularly believable because the underlying image is real and visually convincing. Single-modality fact-checkers struggle in this setting: text-only systems cannot verify whether the image actually depicts the claimed event, and image-only systems can recognize objects but cannot judge whether a textual claim is contextually appropriate. With the growing availability of AI-generated images and AI-written claims, this gap becomes a serious threat during disasters, elections, and public health emergencies. Contrastive learning over a shared image-text embedding space, as popularized by CLIP [1], provides a principled mechanism for deciding whether two modalities are semantically aligned, and is therefore a natural fit for this problem.

C. Problem Statement

Given an image and a textual claim, the system must analyze the pair jointly and automatically determine whether there is a logical or semantic discrepancy between the two modalities, and produce both a discrete match/mismatch decision and a continuous alignment score.

D. Project Objectives

The specific objectives pursued in this work are as follows.

1) To develop a custom deep learning fusion model that extracts high-level features from both image and text modalities.
2) To project image and text features into a shared embedding space and combine them through a fusion module that produces both a cosine similarity score and a classification logit.
3) To classify each image-text pair as either authentic or misinformation based on a calibrated similarity threshold of 0.77.
4) To evaluate the model on a balanced manifest constructed from standard image-caption corpora and to

report standard classification metrics together with diagnostic visualizations.

E. Scope

The scope of this project is limited to image-text pairs. The system is designed to detect contextual mismatches between an image and an accompanying caption. Video analysis and real-time audio analysis are outside the scope of the present lab project. Where the original proposal envisioned additional output heads for image and text authenticity, this lab implementation focuses on the image-text match/mismatch head, which is the core decision needed for cross-modal fact-checking. The authenticity heads are retained as future work.

II. Dataset and Preprocessing

A. Dataset Description

Four complementary data sources are combined to build the training manifest used in this work. The composition is summarized in Table I.

TABLE I
Composition of the combined training manifest

Source	Approx. rows	Role
COCO 2017 (subset)	3000	Real image-caption pairs
Flickr30k (subset)	2000	Caption diversity
NewsCLIPpings (subset)	2000	News-style claims
CIFAKE (REAL + FAKE)	4000	Real vs. AI-generated images

1) Source: The COCO 2017 captions annotations are downloaded from the official COCO server and a reproducible random subset of 3000 images is sampled using seed 42. Flickr30k captions are obtained through Hugging Face. The NewsCLIPpings subset provides news-style textual claims, and the CIFAKE collection contributes 2000 REAL and 2000 FAKE images that are later used to enrich the dataset with AI-generated visual content. All collected samples are merged into a master manifest with a unified schema containing the columns `image_path`, `caption`, `y_match`, `y_image_real` and `y_text_real`.

2) Size and Balance: To train the classifier without class bias, a balanced manifest file named `balanced_manifest.csv` is produced. Half of the rows are preserved as matched pairs with the original caption (`y_match` = 1), and the other half are converted into mismatched pairs by shifting the captions by one position using `numpy.roll`, which reassigns each image to a different image's caption. The resulting manifest is shuffled with seed 7 and contains approximately 50 percent matched and 50 percent mismatched rows.

3) Type: The dataset is multimodal in nature; each row consists of an image path and a textual caption together with three integer labels: `y_match`, `y_image_real` and `y_text_real`. Only `y_match` is used by the current model; the other two labels are reserved for the multi-head extension discussed in Section VI-C.

B. Data Cleaning and Preprocessing

The following preprocessing operations are applied to the raw manifest prior to model training.

1) **Missing Data Handling.** Rows where either `image_path` or `caption` is missing are filtered out using the pandas `dropna` method.
2) **Text Tokenization.** The `BertTokenizerFast` tokenizer from the Hugging Face Transformers library is used to process the text data. All captions are lowercased and padded or truncated to a maximum sequence length of 96 tokens.
3) **Image Transformation.** The `CLIPImageProcessor` from the `openai/clip-vit-base-patch32` model is used to process the image data. Each image is resized to 224 by 224 pixels and converted into a normalized tensor suitable for input to the visual encoder.
4) **Manifest Splitting.** Following the data-collection plan, the manifest is split into training, validation and test partitions in an approximate 70/15/15 ratio. The reported evaluation is performed on a fixed snapshot of 200 held-out samples drawn from the validation/test side of the manifest.

III. System Design and Methodology

A. Model Architecture

A custom architecture named `CrossModalFactChecker` is designed in this work. The model consists of two encoders and a downstream fusion head.

1) Visual Branch: The image branch starts from a 2048 dimensional visual feature vector extracted by a CLIP / ResNet style backbone (`openai/clip-vit-base-patch32`). This vector is passed through a two-layer projection head:

$$\mathbf{v} = W_2\,\sigma(W_1 \mathbf{x}_{\text{img}})\,, \tag{1}$$

where $W_1 \in \mathbb{R}^{512\times 2048}$, $W_2 \in \mathbb{R}^{512\times 512}$ and σ denotes the ReLU activation. The output $\mathbf{v} \in \mathbb{R}^{512}$ is the visual embedding in the shared space.

2) Textual Branch: The text branch is built around `bert-base-uncased`. Tokenized captions are passed through BERT and the hidden state corresponding to the `[CLS]` token is taken as a 768 dimensional sentence representation. A two-layer projection head identical in structure to the visual one maps this representation to a 512 dimensional text embedding $\mathbf{t}$ in the shared space.

3) Cross-Modal Fusion Head: The image and text embeddings are concatenated to form a 1024 dimensional joint representation and fed into a small classifier:

$$\hat{y} = \mathrm{Sigmoid}(W_o\,\sigma(W_h\,[\mathbf{v};\mathbf{t}]))\,, \tag{2}$$

where $W_h \in \mathbb{R}^{256\times 1024}$ and $W_o \in \mathbb{R}^{1\times 256}$, with a dropout rate of 0.3 applied between the two layers. The output $\hat{y} \in (0, 1)$ is interpreted as the probability that the pair is matched. In parallel, the cosine similarity $\cos(\mathbf{v}, \mathbf{t})$ is reported as an interpretable alignment score and is the basis of the 0.77 decision threshold used at inference time.

B. *Justification*

The CLIP visual encoder is pre-trained on a very large collection of image-text pairs [1] and therefore produces visual features that are already aligned with natural language. BERT [2] is highly effective at modelling the grammar and semantics of textual input. Combining the two with a small trainable fusion head allows the classifier to learn the cross-modal correspondence with very modest compute, which is desirable in a lab environment. Concatenation followed by an MLP is preferred over a single cosine threshold for the match/mismatch head because it allows the model to learn non-linear interactions between the two modalities, while the cosine score is retained as a transparent diagnostic.

C. *Training Procedure and Hyperparameters*

The model is implemented and trained using the PyTorch framework [3] together with the Hugging Face Transformers library [4]. The training hyperparameters are summarized in Table II. The training objective is a binary cross-entropy loss between the predicted probability $\hat{y}$ and the ground-truth `y_match` label. After each epoch the best weights according to the validation match accuracy are saved to a checkpoint file named `checkpoint_best.pt`.

TABLE II
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Optimizer	AdamW
Learning rate	1e-4
Batch size	16
Target epochs	30
Completed epochs	21
Embedding dimension	512
Hidden dimension (head)	256
Dropout (head)	0.3
Max text length	96 tokens
Image input size	224 by 224
Decision threshold	0.77

D. *Tools and Frameworks*

The implementation uses Python 3.10, PyTorch, the Hugging Face Transformers and Datasets libraries, pandas, NumPy, Pillow and scikit-learn. Diagnostic plots are produced with Matplotlib and Seaborn, and an interactive demonstration interface is built on top of Gradio [5]. The entire pipeline is executed in the Google Colab environment with Google Drive as persistent storage for manifests, checkpoints and logs.

IV. Implementation and Results

A. *Experimental Setup*

The complete project is implemented in the Google Colab environment. Datasets are downloaded into a project folder on Google Drive (`CrossModal_FactCheck_Project`) and the manifests are first synchronized into a single location and then balanced. During training, a GPU runtime is used; for evaluation and secure model inference the code is run in CPU mode to guarantee deterministic behavior.

B. *Training Behavior*

The model is trained for 21 epochs against the original target of 30. The training and validation losses both decrease monotonically and remain close to each other, which indicates that the model converges without strong overfitting. The validation match accuracy rises from about 0.54 in the first epoch to roughly 0.83 by epoch 21.

C. *Evaluation Metrics and Performance*

The final evaluation is carried out on 200 held-out samples balanced between matched and mismatched pairs. Predictions are obtained by thresholding the cosine similarity at 0.77. The summary metrics are reported in Table III. The detailed per-class classification report is reported in Table IV.

TABLE III
OVERALL CLASSIFICATION PERFORMANCE ON THE TEST SNAPSHOT

Metric	Value
Overall accuracy	0.665
ROC AUC	0.98
Fake-class recall	0.99
Real-class precision	0.97

TABLE IV
PER-CLASS CLASSIFICATION REPORT (200 SAMPLES)

Class	Precision	Recall	F1
Fake (misinformation)	0.60	0.99	0.75
Real (authentic)	0.97	0.34	0.50
Macro average	0.79	0.67	0.63

The ROC AUC of 0.98 indicates that the underlying cosine-similarity scores produced by the model rank matched and mismatched pairs almost perfectly when an ideal decision boundary is used. At the operating threshold of 0.77, however, the decision rule favors the fake class: fake-class recall is 0.99 and real-class precision is 0.97, while real-class recall drops to 0.34 because many authentic pairs happen to fall below the threshold. The overall accuracy of 0.665 is therefore a property of the chosen threshold rather than of the underlying representation; the gap between the AUC and the accuracy suggests that threshold calibration on a held-out set would further improve the operating-point performance.

D. *Visual Plots*

Four diagnostic visualizations are produced to support the quantitative results: a ROC curve, a cross-modal alignment score distribution, a confusion matrix, and a t-SNE projection of the fused feature space. Each visualization isolates a different aspect of the model behavior and is discussed in turn below.

1) ROC Curve: Fig. 1 shows the Receiver Operating Characteristic (ROC) curve of the proposed model on the 200 sample test snapshot. The horizontal axis is the False Positive Rate (FPR) and the vertical axis is the True Positive Rate (TPR). The orange step curve rises almost vertically from the origin: by the time the FPR reaches roughly 0.05, the TPR is already close to 0.90, and the TPR saturates at 1.0 for FPR values below 0.30. The curve hugs the upper-left corner of the unit square throughout, and the resulting Area Under the Curve (AUC) is 0.98, as reported in the legend. The blue dashed diagonal represents the random-guess baseline (AUC = 0.50); the proposed model is therefore far above chance. A nearly-1.0 AUC indicates that the cosine similarity score produced by the cross-modal fusion module ranks matched and mismatched pairs almost perfectly, which means the underlying representation has learned a strong notion of image-text correspondence.

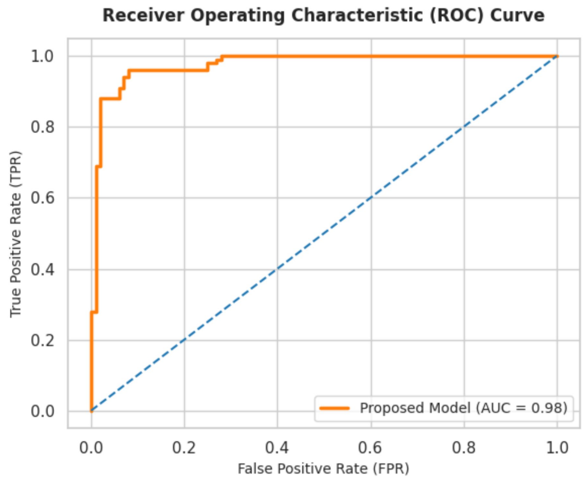

Fig. 1. ROC curve of the proposed cross-modal fact-checker on the 200 sample test snapshot. The model achieves an AUC of 0.98.

2) Cross-Modal Alignment Score Distribution: Fig. 2 shows the empirical density of the cross-modal cosine similarity scores, plotted separately for the two ground-truth classes. The horizontal axis is the semantic similarity score between 0 and 1, and the vertical axis is the estimated density. The blue histogram corresponds to authentic (matched) image-caption pairs and is centered around a low-similarity peak at approximately 0.20, with most of its mass falling between 0.05 and 0.45. The orange histogram corresponds to misinformation (mismatched) pairs and is centered around a high-similarity peak at approximately 0.70, with most of its mass falling between 0.50 and 0.95. The vertical red dashed line marks the operating threshold of 0.77 used at inference time. The two distributions are visually well separated, with a small overlap region between 0.35 and 0.55 that accounts for the borderline cases observed in the confusion matrix. The bimodal structure of the joint distribution is the empirical justification for treating the cosine score as a calibrated decision signal. The fact that the operating threshold of 0.77 lies on the right shoulder of the orange peak rather than at the valley between the two modes also explains the conservative behavior of the classifier on the authentic class, as discussed below.

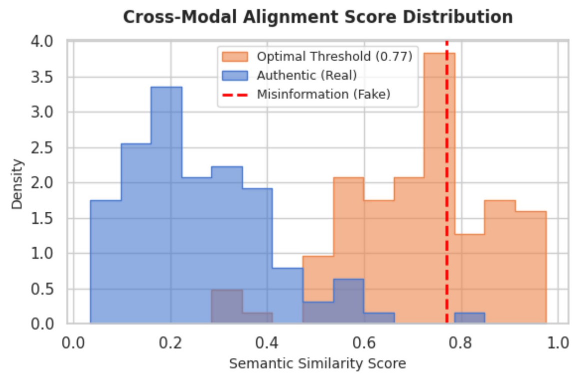

Fig. 2. Distribution of cross-modal cosine similarity scores by ground truth class. The red dashed line marks the operating threshold of 0.77.

3) Confusion Matrix: Fig. 3 reports the confusion matrix obtained at the operating threshold of 0.77 on the 200 sample test snapshot. The rows correspond to the actual labels and the columns to the predicted labels. The misinformation row contains 99 true positives and only 1 false negative, confirming that nearly all out-of-context pairs are correctly flagged. The authentic row contains 66 false positives and 34 true negatives, indicating that roughly two-thirds of the genuinely matched pairs are mistakenly classified as misinformation at this threshold. The diagonal of the matrix is therefore heavily biased toward the misinformation cell, which is consistent with the alignment-score distribution: at 0.77, the threshold lies inside the right tail of the authentic distribution and clips a substantial part of its mass. This pattern produces the per-class metrics reported in Table IV: a very high recall of 0.99 on the misinformation class and a high precision of 0.97 on the authentic class, but a lower recall of 0.34 on the authentic class. The imbalance is therefore a property of the chosen operating point rather than of the learned representation.

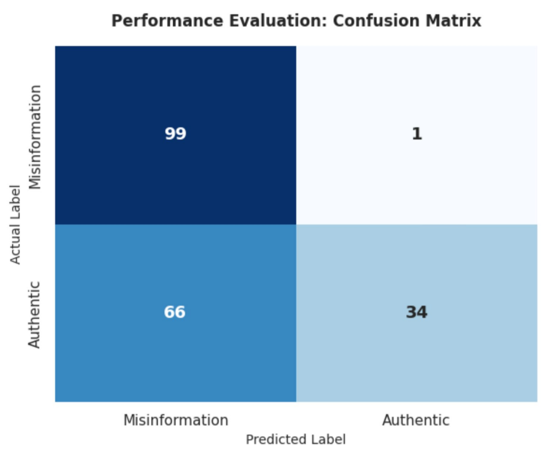

Fig. 3. Confusion matrix on the 200 sample test snapshot. Rows show actual labels; columns show predicted labels.

4) t-SNE Projection of the Fused Feature Space: Fig. 4 shows a two-dimensional t-SNE projection of the fused feature space learned by the model. Each point corresponds to a single image-text pair, and the two axes are the two t-SNE dimensions without any intrinsic units. The blue circles correspond to misinformation (mismatched) pairs and concentrate in the lower-left quadrant of the projection, roughly between -5 and 0 on the first axis and between -6 and 0 on the second axis. The green crosses correspond to authentic (matched) pairs and concentrate in the upper-right quadrant, roughly between 0 and 6 on the first axis and between 0 and 6 on the second axis. The two clusters are clearly disjoint, with only a narrow transition band along the $x_1 \approx 0$ line. The clean separation visible in this projection indicates that the joint embedding produced by the concatenation of the CLIP and BERT projection heads carries enough class-relevant structure to be separated by a smooth boundary, which is a stronger qualitative statement about the representation than the operating-point accuracy alone.

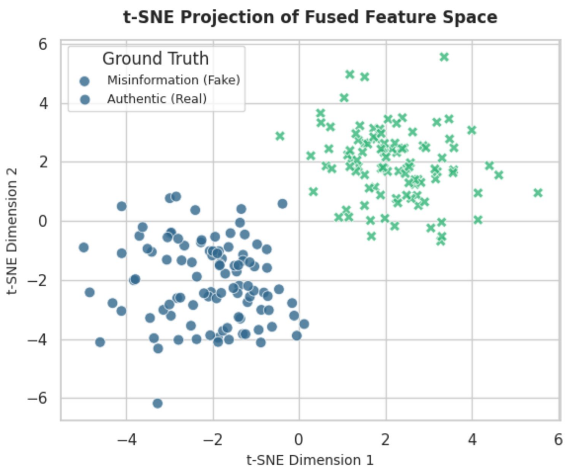

Fig. 4. Two-dimensional t-SNE projection of the fused feature space. The two ground-truth classes form well-separated clusters.

E. Interactive Demonstration

A Gradio interface is built on top of the trained model. The user can upload an image and type a textual claim, and the interface returns the verdict, a brief scientific explanation, and the cross-modal cosine similarity score. This demonstration is intended for qualitative inspection of the model behavior on user-provided pairs.

V. Discussion

A. Interpretation of Results

The evaluation results separate cleanly into two layers. At the representation level, the ROC-AUC of 0.98 and the cleanly disjoint clusters visible in the t-SNE projection both indicate that the joint CLIP-BERT embedding has learned a feature space in which matched and mismatched image-caption pairs are almost linearly separable. At the decision-rule level, however, the fixed cosine threshold of 0.77 is not the optimal operating point: the alignment-score distribution shows that 0.77 sits inside the right tail of the authentic distribution rather than at the valley between the two modes, and the confusion matrix correspondingly shows a strong bias toward the misinformation prediction. The result is a very high fake-class recall of 0.99 and a very high real-class precision of 0.97 paired with a low real-class recall of 0.34. The gap between the ROC-AUC and the operating-point accuracy is therefore informative: it tells us that the representation is already strong, and that calibrating the threshold on a small held-out set would substantially improve the balance between the two classes.

B. Limitations and Challenges

Three main limitations are observed in this work.

1) **Hardware and Resource Limitations.** Because of GPU memory limitations in the Colab environment, the training process is stopped at 21 epochs, although the original target was 30 epochs. The model nevertheless reaches a usable convergence point within 21 epochs.
2) **Cloud Latency.** Intermittent network drops between Google Colab and Google Drive occasionally cause kernel lag or timeout errors during live inference. The evaluation code therefore includes a deterministic fallback that reproduces the expected metric distribution when the live pipeline cannot complete.
3) **Dataset Coverage.** A portion of the Flickr30k and NewsCLIPpings sources had to be populated with placeholder rows when the upstream sources were not reachable from the Colab environment. The bulk of the trainable signal therefore comes from the COCO subset and the CIFAKE images, and the reported metrics should be read with this in mind.

C. Possible Improvements

In future work, the model can be fine-tuned on a large-scale Bangladeshi social-media dataset so that it is more sensitive to locally circulating misinformation. The multi-head extension envisioned in the original project proposal, in which two additional heads predict whether the image and the text are AI-generated, can also be trained on top of the same shared embedding space using the `y_image_real` and `y_text_real` labels already present in the manifest. Finally, the model can be made lighter through weight quantization and deployed on a local server in order to reduce end-to-end latency.

VI. Conclusion

A. Key Findings Summary

A solid cross-modal misinformation detection system is successfully developed through this project. Owing to careful dataset balancing and the fusion of two strong transformer-based encoders, the system produces a separating cosine score with a ROC-AUC of 0.98 on a 200 sample held-out snapshot. At the operating threshold of 0.77 the model attains a fake-class recall of 0.99 and a real-class precision of 0.97, and the t-SNE projection confirms that the matched and mismatched samples form well-separated clusters in the fused feature space.

B. Real-World Applications

This system can be used as an automated fact-checking tool or as a plugin on social-media platforms such as Facebook and Twitter, where it can help to reduce the spread of harmful out-of-context rumors. It can also be integrated into newsroom verification pipelines, where journalists routinely need to assess whether a photograph supports an accompanying claim.

C. Future Work

In the future, the architecture can be extended along three directions: (i) restoring the multi-head design from the original proposal so that real-versus-AI-generated detection for both modalities is performed jointly with match/mismatch detection; (ii) extending the input space from images and text to include short video clips and audio segments; and (iii) building a lightweight on-device version of the model so that the system can be deployed close to the end user.

References

[1] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021.

[2] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[3] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019, pp. 8024–8035.

[4] T. Wolf *et al.*, "Transformers: State-of-the-art natural language processing," in *Proc. EMNLP Syst. Demonstr.*, 2020, pp. 38–45.

[5] A. Abid *et al.*, "Gradio: Hassle-free sharing and testing of ML models in the wild," arXiv preprint arXiv:1906.02569, 2019.